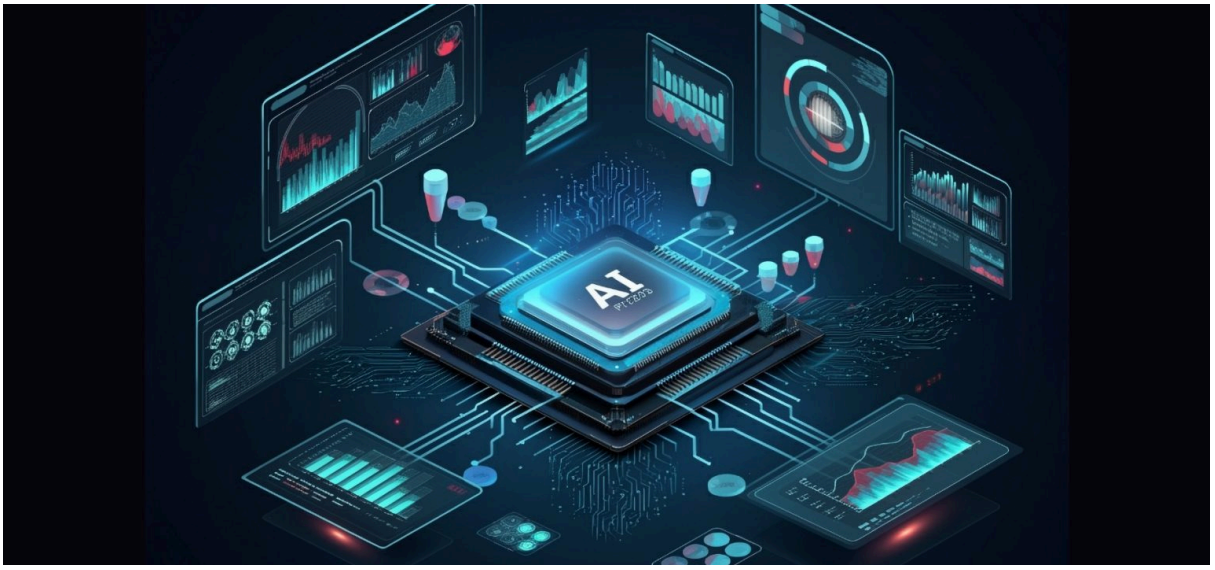


Astra-OS

AI-Native Operating System

Product Vision

> **Build an AI-native operating environment that understands user intent, automatically optimizes hardware resources, and enables no-code AI development on everyday computers.**

Problem Statement

Current operating systems are designed for **general-purpose computing**, not AI workloads.

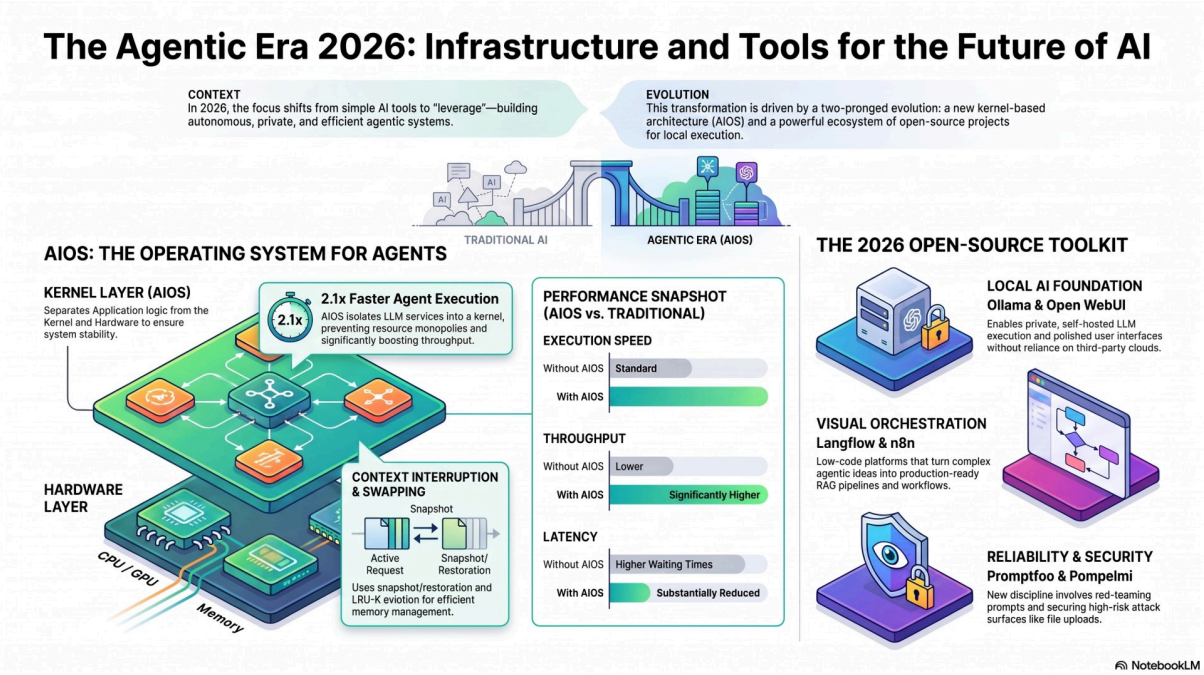
Users have to:

- Install multiple tools
- Configure ML environments
- Write code
- Manage CPU/GPU resources
- Tune memory

- Troubleshoot failures

This makes AI development difficult, especially on low-resource hardware.

Solution

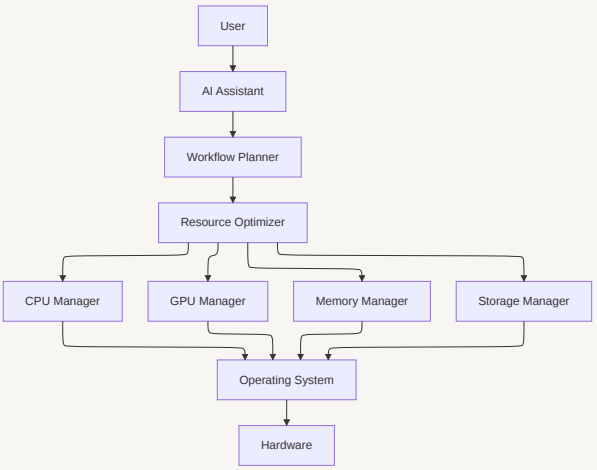


Our AI-powered operating system introduces an **AI Orchestration Layer** that sits above the traditional operating system.

Instead of users managing the computer,

the AI manages the computer for the user.

System Architecture



Traditional OS vs Proposed AI OS



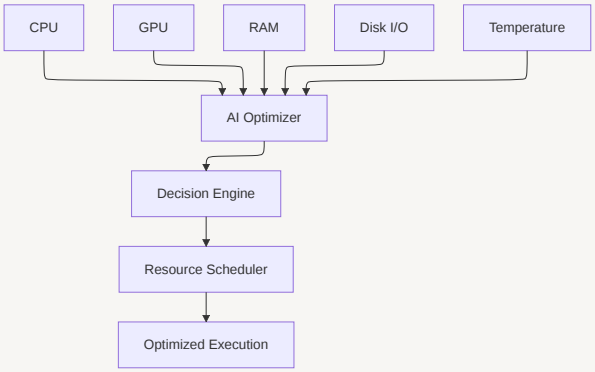
Proposed System



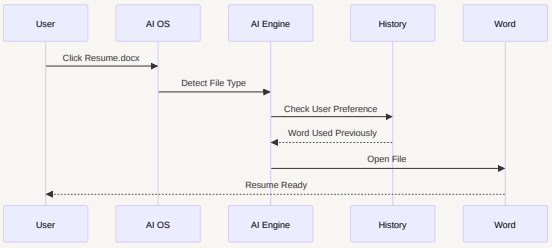
AI Training Workflow



Resource Optimization Flow



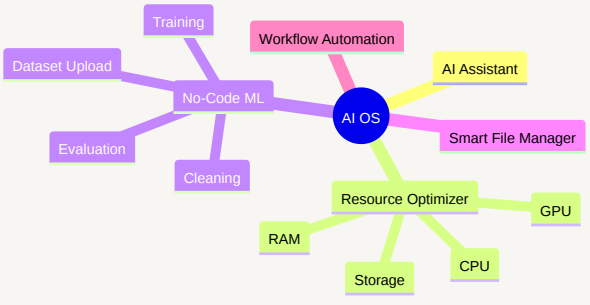
Smart File Opening



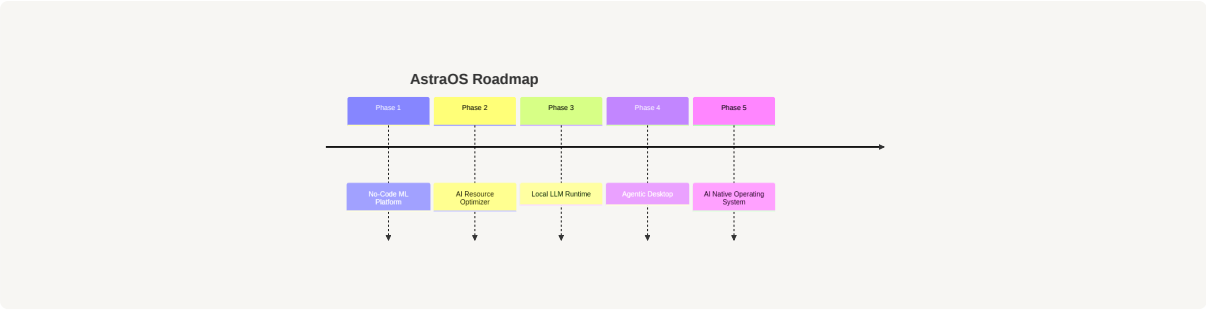
No-Code ML Workflow



Functional Modules



Future Roadmap



Research Papers

IEEE

1. AI-Empowered Fog/Edge Resource Management for IoT Applications

IEEE Communications Surveys & Tutorials (2024)

https://ieeexplore.ieee.org/document/10335918

Focus

- AI scheduling
- Resource optimization
- CPU allocation

2. A Comprehensive Review of AI Techniques for Resource Management in Fog Computing

IEEE Access (2024)

https://ieeexplore.ieee.org/document/10627948

Focus

- AI scheduling
- Intelligent orchestration
- Resource management

3. Edge-LLM: A Collaborative Framework for Large Language Model Serving in Edge Computing

IEEE ICWS (2024)

https://ieeexplore.ieee.org/document/10707514

Focus

- Edge AI
- LLM optimization
- Low-resource inference

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4. Distributed AI in Zero-Touch Provisioning for Edge Networks

IEEE Computer (2024)

https://ieeexplore.ieee.org/document/1046168

Focus

- Autonomous resource management
- AI automation

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5. Decentralized Scheduling and Dynamic Pricing for Edge Computing

IEEE/ACM Transactions on Networking

https://ieeexplore.ieee.org/document/9903403

Focus

- Scheduling
- Distributed optimization

Other Highly Relevant Papers

AIOS: LLM Agent Operating System

https://arxiv.org/abs/2403.16971

Closest research to your idea.

Introduces

- AI kernel
- Agent scheduler
- Context management
- Memory management
- Tool management

Achieves **up to 2.1× faster execution** for serving agent workloads in their evaluation. (arXiv)

LLM as OS, Agents as Apps

https://arxiv.org/abs/2312.03815

Introduces the concept

> LLM becomes the Operating System

Agents become Applications.

(arXiv)

Semantic File System for AIOS

https://arxiv.org/abs/2410.11843

Very relevant to your idea of opening files intelligently instead of relying only on filenames and folders. (arXiv)

AgentRM: An OS-Inspired Resource Manager for LLM Agent Systems

https://arxiv.org/abs/2603.13110

Shows how operating-system concepts like scheduling and memory management can improve agent execution. (arXiv)

USP

- AI-first operating environment
- No-code machine learning
- Intelligent CPU/GPU/RAM optimization

- Goal-based computing instead of application-based computing
- Smart file and workflow management
- AI-assisted execution on low-resource hardware

Future Scope

- Autonomous AI scheduling
- Local LLM execution
- Multi-agent collaboration
- AI desktop assistant
- Intelligent memory optimization
- AI-powered developer environment
- Enterprise AI workstation platform

This structure is suitable for a **final-year engineering project, hackathon, startup pitch, or research proposal**, and the Mermaid diagrams can be rendered directly in Markdown-supported tools.